

TIC - Big Project Internship Summer26

Product Requirements Document (PRD)

Product Name: Cardly – AI-Powered Business Card Scanner & Smart Contact Forge

Version: 1.0

Date: May 2026

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Status: Ready for team kick-off

1. Product Overview

Cardly is a mobile app that turns the traditional Vietnamese networking ritual of exchanging paper business cards into a seamless, intelligent, and future-proof digital experience.

Users simply open the app and scan or photograph a business card. The AI instantly extracts all key information, creates a rich contact record, and then uses an intelligent agent to enrich the contact with additional insights from LinkedIn, company website, Facebook, etc. Every contact is automatically tagged with context (event name, date, location) and enriched with a short brief, keywords, and highlights about the person and their company.

The app also generates a beautiful public digital business card page (Linktree-style) for the user themselves, making it easy to share their own profile.

Core Value Proposition

"Scan once. Own forever. Know more."

2. Problem Statement

In Vietnam, business card exchange is still the dominant way to network at conferences, events, meetups, and meetings. However:

- Paper cards get lost, damaged, or forgotten.
- Manually typing information into phone contacts or CRM is time-consuming and error-prone.
- Users have no easy way to remember context (who they met, where, when, and why they are important).
- There is no intelligent follow-up layer (brief about the person/company, keywords for future search, highlights).
- Users want a single beautiful digital version of themselves (like Linktree) to share instead of handing out paper cards.

This creates friction in networking and wastes valuable relationship-building opportunities.

3. Goals & Objectives

Primary Goals

- Make business card collection instant, accurate, and effortless.
- Turn every scanned card into a rich, searchable, and context-aware contact.
- Provide users with a professional digital business card (public profile) they can share instantly.
- Build the foundation for a full smart contact management platform.

Success Metrics (MVP)

- 90%+ accuracy on name, phone, email, company, position extraction.
- Average time to create a contact < 8 seconds.
- User satisfaction (NPS $\geq$ 70 after first 10 scans).
- 80% of users create at least one digital business card profile.

4. Target Audience

Primary Users

- Professionals, entrepreneurs, salespeople, startup founders, and students in Vietnam who attend networking events, conferences, meetups, and business

meetings (age 22–45).

Secondary Users

- Companies/teams that want team-wide contact management and digital business card system.

5. Core Features (MVP)

5.1 Core Scanning & Extraction

- Open camera → scan or take photo of business card (supports Vietnamese + English text).
- AI extracts: Full name, position, company, phone, email, website, LinkedIn/Facebook (if present), address, QR code if any.
- Auto-creates contact record in app database.

5.2 Intelligent Enrichment Agent

- After extraction, an agent automatically:
 - Searches LinkedIn, company website, and Facebook (public data only).
 - Generates:
 - Short professional brief (2–4 sentences)
 - Key keywords/tags (e.g., "AI startup", "renewable energy", "sales director")
 - Highlights (e.g., "Recently raised Series A", "Expert in solar energy policy")
- User can review/edit the enrichment before saving.

5.3 Smart Tagging & Context

- Auto-tags every contact with:
 - Date & time of scan
 - Event name / location (user can select from list or input manually)
 - Source ("Conference – TechDay 2026", "Coffee meetup", etc.)
- Manual tagging and notes supported.

5.4 Manual Contact Creation

- Full manual form to add contacts without scanning.

5.5 My Digital Business Card (Public Profile)

- User creates one beautiful, shareable public page (Linktree-style):
 - Profile photo, name, title, company
 - Contact buttons (phone, email, WhatsApp, Zalo, LinkedIn)
 - Bio + highlights
 - Custom URL (e.g., cardly.me/yourname)
- Easy sharing via QR code, link, or message.

6. Future / Phase 2 Features (Brainstorm)

- Sync with phone contacts (iOS & Android)
- Two-way sync with Gmail / Google Contacts / iCloud
- Export to Notion, Airtable, HubSpot, or custom CRM via API
- Team workspaces (company shared contacts)
- Smart reminders ("Follow up with Mr. Anh – met at Vietnam Tech Summit")
- AI-generated follow-up message templates
- Analytics dashboard ("Your networking activity")
- NFC business card mode (tap phone to share digital card)
- Bulk scan multiple cards at events

7. User Flows (High-Level)

Main Flow – Scan a Card

1. Open app → Tap "Scan Card"
2. Camera opens → User takes photo or scans
3. AI processes → Shows extracted data + confidence
4. Enrichment agent runs (progress indicator)

5. User reviews/edits brief, keywords, tags
6. Save → Contact created with full context

My Digital Card Flow

1. Profile → "My Digital Card"
2. Customize bio, photo, links
3. Generate public URL + QR code
4. Share instantly

8. Technical Considerations

- **Mobile:** Flutter or React Native (cross-platform)
- **Backend:** GCP (Vertex AI for OCR + enrichment models)
- **AI Stack:**
 - OCR + extraction: Fine-tuned open-source model on HF (e.g., Donut, LayoutLM, or Qwen2-VL)
 - Enrichment agent: Vertex AI Agent Builder / ADK + small open LLM
- **Database:** Firestore or PostgreSQL + vector search for future semantic search
- **Storage:** Cloud Storage for original card images (private)

9. Non-Functional Requirements

- Privacy-first: All data encrypted, user owns their contacts
- Offline support for scanning (sync when online)
- Fast processing (< 10 seconds end-to-end)
- Support Vietnamese + English text
- High accuracy on Vietnamese business cards

10. Roadmap

Phase 1 – MVP (8–10 weeks)

- Core scanning + extraction

- Auto contact creation
- Basic enrichment agent
- Manual contact
- My Digital Business Card (simple version)

Phase 2 (next 6–8 weeks)

- Advanced enrichment (LinkedIn/website scraping)
- Smart tagging
- Full digital card customization
- Basic sync features

Phase 3

- Team features, CRM integrations, advanced AI capabilities